MIL-F-17548B(SHIPS)
23 January 1962
SUPERSEDING
MIL-F-17548A(SHIPS)
10 March 1954

MILITARY SPECIFICATION

FLAME ARRESTER, VENTILATION-EXHAUST

(NAVAL SHIPBOARD USE)

1. SCOPE

1.1 This specification covers flame arresters for preventing the propagation of flame through exhaust ventilating ducts which may contain flammable mixtures.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of the invitation for bids, form a part of this specification, to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-901 - Shockproof Equipment, Class HI, (High-Impact) Shipboard Application, Tests for.

MIL-F-16552 - Filters, Air Conditioning, Cleanable, Impingement.

MIL-P-17545 - Primer Coating. Steel Ship Maintenance, Alkyd-Red Lead Type, Formula No. 116.

STANDARDS

MILITARY

MIL-STD-278 - Welding and Allied Processes for Machinery for Ships of the United States Navy

DRAWINGS

BUREAU OF SHIPS

S3801-1231181 - Hull Type Drawing for Flame Arrester and Filter assembly.

(Copies of specifications, standards, drawings and publications required in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

A-93 - Sheets, Coils and Cut Lengths, Zinc Coated (Galvanized) Iron or Steel

A-164 - Zinc on steel, Electro-Deposited Coating of

A-386 - Zinc Coating (Hot Dip) on Assembled Steel Products

A-415 - Hot-Rolled Carbon Steel Sheets, Commercial Quality

B-6 - Slab Zinc (Spelter)

B-201 - Chromate Finishes on Electro-Deposited Zinc, Hot-Dipped Galvanized and Zinc Die-Coat Surfaces

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia 3, Pennsylvania.)

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N. Y.)

3. REQUIREMENTS

3.1 Qualification (size 38 only). - The size 38 flame arrester assembly furnished under this specification shall be a product which has been tested and passed the qualification tests specified herein and has been listed on or approved for listing on the applicable qualified products list.

3.2 Preproduction sample (except size 38). - Prior to production, one flame arrester of each size

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(except size 38) shall be submitted for tests as specified in 4.3.

3.3 Materials.- Materials not covered by this specification shall be of a high grade, free from defects and suitable for purpose intended. Aluminum, aluminum alloys, cast iron or malleable iron, shall not be used in any part of the flame arrester assembly.

3.4 Design.- The flame arrester assembly shall consist of a watertight casing, an arresting media which shall prevent the passage of a flame from one side of the media to the other side when tested as specified herein, and cleanable high velocity impingement type air filters.

3.4.1 The flame arrester shall be intended for operation under shipboard condition of shock.

3.4.2 Sizes.- Sizes of flame arrester assemblies shall be limited to those on Drawing S3801-1231181.

3.4.2.1 Physical dimensions and tolerances.- The physical dimensions and tolerances of the flame arrester assembly shall conform to Drawing S3801-1231181 for sizes specified.

3.4.3 Casing.- The flame arrester assembly casing shall be made of galvanized steel equal to that of ASTM Publication-A-93 with commercial thickness zinc coating or black steel equal to that of ASTM Publication -A-415.

3.4.3.1 Casing and other parts made of black steel shall be galvanized by the hot-dip process in accordance with ASTM Publication-A-386 with the spelter conforming to grade 5 of ASTM Publication-B-6, or by electroplating with zinc in accordance with type LS of ASTM Publication-A-164 followed by a chromate treatment conforming to ASTM Publication-B-201.

3.4.4 Flame arresting media.- The flame arresting media shall be constructed of galvanized steel or bronze and shall be permanently secured in the casing in a manner to prevent passing of a flame between the media and the casing.

3.4.5 Air filters.- Air filters shall be provided in the air entering side of the casing to protect the arresting media from dust. The air filters shall be in accordance with Specification MIL-F-16552.

3.4.5.1 Means, such as flat springs, shall be provided to hold the air filters firmly within the casing.

3.4.6 Access cover.- A watertight access cover shall be provided in the casing for servicing the air filters. The access cover gasket shall be of cork, neoprene or a 3 ply butyl rubber tape, fabric reinforced and pressure sensitive on both sides.

3.4.7 Pressure loss (resistance to airflow).- The resistance to airflow through the flame arrester, with air filters removed, shall not exceed the values shown in table I.

3.5 Welding.- The surfaces of all parts to be welded shall be free from rust, scale, paint, grease and other foreign matter. All welding shall be at least equal to that required by Standard MIL-STD-278.

3.5.1 Where zinc-coated steel is used for fabricating parts, the metallic zinc shall be removed from all points and surfaces on which welds are to be deposited and for a distance of 1/2 inch from the expected toes of the welds. In areas where the metallic zinc cannot be removed and it is necessary to weld over the zinc-coated surface, electrode AWS-E-6010 shall be used.

3.6 Finish.- Surfaces to be painted shall be cleaned to remove dirt, grease, corrosion products and other surface contaminants prior to application of primer coating.

3.6.1 Unless otherwise specified herein, all zinc-coated surfaces including welds and areas damaged by welding, grinding, shearing, punching or any fabricating process shall receive a thorough wash and a hot phosphate acid treatment prior to applying two coats of primer conforming to Specification MIL-P-17545.

3.6.2 Flame arresting media and air filters shall not be painted.

3.7 Fasteners.- All bolts, nuts, screws, and other such fastenings as may be used shall be of a corrosion-resisting material suitable for intended application and shall be of a recognized commercial standard shape, size, length and tolerances. Unless otherwise specified (see 6.1), bolts, nuts, and machine screws shall have unified coarse-thread series threads.

3.8 Drawings.- Drawings delineating the equipment shall be furnished in accordance with class C of Specification MIL-D-963 and as specified herein. Drawings are not to be titled to show application to particular ships or purchase orders.

3.8.1 Assembly drawings.- Assembly drawings shall show clearly the details of design, construction and identify each part and its location. Parts shall be identified in accordance with the numbers used in the list of material. The drawing shall show overall dimensions, mounting dimensions, attachment and

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connection dimensions, and tolerances. Such additional information, not required herein, shall be included on the drawing to reflect compliance with the specification requirements for the equipment.

Table I - Values

Face velocity	Pressure loss
Feet per minute	Inches (water gage)
400	0.27
600	.60
800	1.10

3.8.2 Certification data.- Certification data shall be furnished and shall conform to class B of Specification MIL-D-963, with approval by the bureau or agency concerned.

3.9 Identification plates.- Identification plates and other designating marking plates may be etched, engraved, or cast, using either monel, brass, corrosion-resisting steel or aluminum alloy or may be of anodized aluminum photographic process. These plates shall be secured to the outside of the casing in such a manner as not to destroy the watertight integrity of the casing.

3.9.1 Each flame arrester assembly shall be provided with an identification plate not to exceed 3 by 3 inches and shall contain the following information:

Flame Arrester, Ventilation-Exhaust
Size
Federal stock number (FSN)
Component identification number (CID)
Contract number
Manufacturer's name
Manufacturer's address

3.9.2 Each flame arrester assembly shall be provided with an arrow or similar means indicating the direction of airflow.

3.9.3 Aluminum photographic process.- Aluminum anodized plates shall have the photosensitive silver compounds imbedded within the oxide layer. After photographic processing, the image shall be sealed in the oxide layer by suitable chemical treatment of the layer. Stamping of additional marking information will not normally be permitted on metal plates.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Qualification tests (size 38 only).[1/]- Qualification tests shall be conducted at a laboratory satisfactory to the Bureau of Ships. Tests shall be conducted as specified in 4.2.1.

4.2.1 Qualification tests shall be conducted on one size 38 flame arrester assembly and shall consist of tests specified in 4.7.

4.3 Preproduction tests (except size 38).- Production of items under the contract or order shall not proceed until one flame arrester of each size (except size 38) covered in the contract or order shall have successfully passed the tests specified in 4.7.2 to 4.7.4, inclusive. The contractor may utilize his own or any other laboratory satisfactory to the Bureau of Ships, for the tests to be conducted.

4.3.1 Unless otherwise specified in the contract or order, preproduction tests will not be required on sizes of flame arresters that are identical to those which have successfully passed the preproduction tests within a period of twelve months immediately preceding the award of the contract.

4.4 Sampling for lot acceptance.-

4.4.1 Lot.- All flame arresters of the same size shall be considered a lot.

4.4.2 Sampling for examination and test.- Sample flame arresters shall be selected at random from each lot in accordance with table II for the examination specified in 4.5 and the tests specified in 4.6. The Acceptance Quality Level (AQL) shall be 1.5 percent.

4.5 Examination.- Each of the sample flame arresters selected in accordance with 4.4.2 shall be examined and measured to verify compliance with the requirements of this specification, not involving tests. Examination shall be conducted as specified

[1/] Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.2 and 6.3).

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in table III. Any flame arrester in the sample containing one or more defects shall be rejected and if the number of defective flame arresters in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

Table II - Sampling for examination and tests

Number of arresters in lot	Number of arresters to be inspected	Acceptance number (defectives)	Rejection number (defectives)
Up to 8	All	-----	-----
9 to 15	7	0	1
16 to 25	10	0	1
26 to 65	15	0	1
66 to 110	20	1	2
111 to 180	25	1	2
181 to 300	35	2	3

Table III - Classification of defects

Categories	Defects
Critical	None defined
Major	
101	Size not as specified
102	Incomplete, component parts missing
103	Evidence of unauthorized material used
104	Casing or enclosure material thickness; inside and outside dimensions non-conforming
105	Welding (as applicable) incomplete, not free of cracks, nonfusion, heavy porosity, heavy undercut, slag inclusions or not smooth as specified
106	Casing or enclosure drilled or punched holes non-conforming; holes missing, location and size not as required.
107	Provisions not provided to hold filter assembly springs in place; springs non-conforming. not located as required.
108	Casing or enclosure hardware (for example, wingnuts, lockwashers, bolts and nuts) not provided, size and length non-conforming.
109	Casing surface not free of cracks, splits, tears, burrs, and sharp edges, or not clean prior to galvanizing or electroplating with zinc as specified.
110	Casing and arresting media coating incomplete, uncoated areas (bare spots), not free of sharp projections.
111	Air filter missing, defective, or not as specified.
112	Access cover and gasket missing, defective, or not as specified.
113	Painting (as applicable) nonconforming.
114	Marking, manufacturer's name or trade mark, direction of airflow arrows missing, not permanent, incorrect or illegible.

4.6 Tests. - Each sample flame arrester selected in accordance with 4.4.2 shall be tested as specified in 4.7.2 and 4.7.4. Any flame arrester in the sample that fails to pass either test shall be rejected and if the number that is rejected in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.6.1 Possible test failures. - Possible test failures are defined as follows:

(a) Shock - Evidence of fragmentation or missile effect of parts, deformation of parts that will permit passage of a flame or destroy the watertight integrity of the casing. (see 4.7.1).

(b) Pressure loss - Pressure drop of flame arrester with air filters removed exceeds the allowable specified. (see 4.7.2).

(c) Flame resistance - Failure to prevent passage of flame from one side of the arrester to the other (see 4.7.3).

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(d) Watertightness - Evidence of leakage when casing or enclosure is subjected to specified hydrostatic test (see 4.7.4).

4.7 Test procedures. -

4.7.1 High impact shock test. - The test shall be as prescribed for type A, grade I of Specification MIL-S-901. The snock test shall be conducted prior to the flame resistance test (see 4.7.3) and watertightness test (see 4.7.4).

4.7.2 Pressure loss test. - The pressure drop in the air stream across the flame arrester, with air filter removed, shall be determined at 400, 600, and 800 feet per minute (fpm) air velocities.

4.7.3 Flame resistance test. - The flame resistance test shall determine that the flame arrester will prevent the passage of flame from a combustible mixture of gasoline vapor and air on one side of the arrester to a similar mixture on the other side of the arrester when the arrester is installed in a section of ventilation duct. The air filters shall be removed from the casing for this test.

4.7.3.1 Test arrangement. -, The test arrangement shall be as shown on figure 1 and as follows:

(a) Transitions shall be symmetrical with maximum included angle of 15 degrees.
(b) Arrester shall be installed with direction of air flow as shown on figure 1.
(c) Flanged connections shall be used.
(d) Duct and transition pieces shall be made from steel of thickness and dimensions shown in table IV.
(e) Ends of ducts shall be fitted with loose fitting plugs that will be blown out by force of the explosion when mixture is ignited.
(f) An orifice shall be fitted on ignition end of the 25 foot duct. The orifice shall be such to raise the air flow pressure drop of the whole set-up including arrester to 3 inches of water.
(g) White gasoline or equal shall be used having an octane rating of 55 or higher.
(h) Ductwork shall be completely filled with gasoline and air in proper proportions and shall be distributed evenly within the ductwork. The arrester shall be tested over the entire combustible range of the mixture.

4.7.3.2 Explosive mixture. - Tests shall be conducted with various gasoline-air vapor mixtures over the entire explosive range using white gasoline or an equal having an octane rating of 55 or higher.

4.7.3.3 Filling with vapor. - Ends of duct shall be fitted with loose fitting plugs that will be blown out by the force of the explosion when mixture is ignited. These plugs are to maintain the concentration of explosive mixture until the mixture is ignited. A hose connection shall be fitted to the 5-foot duct for filling purposes as shown on figure 1, Carburated pure gasoline shall be fed into the test arrangement until the arrangement is uniformly filled throughout.

4.7.3.4 Ignition. - After the ductwork has been filled evenly with the explosive mixture and the filling hose has been removed, the mixture shall be ignited by means of a spark plug located 6 inches from the end of the 25-foot duct as shown on figure 1.

4.7.4 Watertightness test. - The ends of the flame arrester assembly shall be blanked off and the casing subjected to a hydrostatic test of 10 pounds per square inch gage (psig) for a period of not less than 10 minutes.

4.7.5 Any change in design or fabrication of the flame arrester which may change the shock resistance, air pressure drop, flame resistance, or watertightness shall necessitate a repetition of all of these tests.

5. PREPRARATION FOR DELIVERY

5.1 Packaging. - Flame arresters shall be packaged in accordance with the manufacturer's commercial practice.

5.2 Packing. - Unless otherwise specified in the contract or order, the flame arresters shall be packed for shipment in a manner which will insure acceptance by the carrier and safe delivery to destination. Shipping containers or method of packing shall comply with the Uniform Freight Classification Rules or other regulations as applicable to the mode of transportation.

5.3 Special preservation, packaging or packing. - Where special preservation, packaging, or packing is required because of particular shipment or storage requirements, detail requirements shall be as specified (see 6.1).

5.4 Marking. -

5.4.1 Packages. - Unless otherwise specified in the contract or order, each package shall be marked to show the stock number, nomenclature, size, contract number, name and address of manufacturer of packaged flame arrester. Any additional marking requirements shall be as specified (see 6.1).

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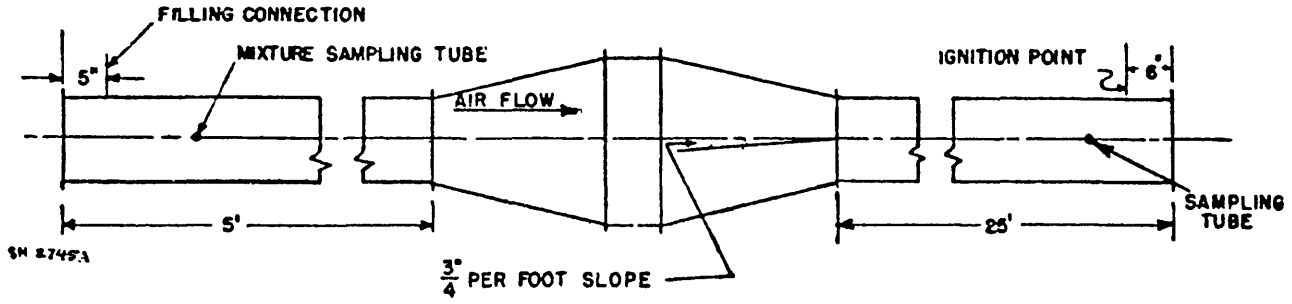


Figure 1 - Test set-up for flame arrester

Table IV - Thickness and dimensions of steel for duct and transition pieces

Size	USS gage	Duct cross section
		Inches
31	26	4 by 4
32	24	5-1/2 by 5-1/2
33	24	7 by 7
34	22	8-1/2 by 8-1/2
35	22	10 by 10
36	20	13 by 13
37	18	16 by 16
38	18	21 by 21

5.4.2 Shipping containers. - Unless otherwise specified (see 6.1), shipping containers shall be marked to show stock numbers, nomenclature, quantity and unit, gross weight and cube, contract number, and the complete address of the consignor and the consignee. Any additional marking requirements shall be as specified in the contract or order.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title number and date of this specification.
(b) If bolts, nuts and machine screws should be other than unified-course-thread series threads (see 3.7).
(c) Detail requirements if special preservation, packaging or packing is required (see 5.3).
(d) Marking of packages and shipping containers (see 5.4.1 and 5.4.2).

6.2 With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List QPL 17548, attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Bureau of Ships, Department of the Navy, Washington 25, D. C., and the information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.3).

6.3 Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise

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SPECIFICATION ANALYSIS SHEET
NAVSHIPS-4863 (8-61)

BUDGET BU. NO. 95-R309

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Bureau of Ships.

This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this form will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to Bureau of Ships, Specifications and Standardization Branch, Washington 25, D.C.

SPECIFICATION

ORGANIZATION	CITY	STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   a. GIVE PARAGRAPH NUMBER AND WORDING

   b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID.

3. IS THE SPECIFICATION RESTRICTIVE? ☐ YES ☐ NO IF THE ANSWER IS "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification.) PLACE THIS FORM AND PAPERS IN AN ENVELOPE AND SEND TO THE BUREAU.

SUBMITTED BY: (Print name and activity)	DATE

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